

Important information. Please read carefully.

Erysol[®]

For Oral Suspension

Composition

Each 5ml when reconstituted contains Erythromycin Ethylsuccinate equivalent to Erythromycin activity 200 mg.

Pharmacodynamics

Erythromycin is a macrolide antibiotic produced by a strain of *Streptomyces erythraeus*. It has a broad spectrum of activity against gram-positive and to a lesser extent some gram-negative bacteria. Erythromycin acts by reversibly binding to 50S subunit of ribosome, inhibiting protein synthesis and hence inhibit bacterial cell growth. It is active against gram positive *Streptococcus pyogenes*, *Streptococcus pneumoniae*, *Staphylococcus aureus*, *Bacillus anthracis*, *Corynebacterium diphtheriae*, *Erysipelothrix rhusiopathiae*, *Listeria monocytogenes*. Anaerobic *Clostridium* spp., *Propionibacterium acnes*, *Nocardia* spp. are also susceptible. Gram negative bacteria includes *Neisseria meningitidis*, *Neisseria gonorrhoeae*, *Moraxella catarrhalis* (*Branhamella catarrhalis*), *Bordetella* spp., some *Brucella* strains, *Flavobacterium* spp., *Legionella* spp., *Haemophilus ducreyi*, *Haemophilus influenzae*, *Helicobacter pylori*, *Campylobacter jejuni* are susceptible. Other organisms sensitive to erythromycin includes Actinomycetes, Chlamydiaceae, rickettsias, spirochaetes such as *Treponema pallidum* and *Borrelia burgdorferi* and some mycoplasma such as *Mycoplasma pneumoniae*, and some opportunistic mycobacteria: *Mycobacterium scrofulaceum* and *M.kansasii*.

Pharmacokinetics

Food has little effect on the absorption of erythromycin ethylsuccinate. Peak plasma concentrations generally occur between 1 and 4 hours range from 0.3 to 1.0 micrograms/mL after administration of 250mg of erythromycin base. Erythromycin is widely distributed throughout body tissues although it does not cross blood brain barrier well. Erythromycin crosses the placenta and distributed into breast milk. 70 to 75% of erythromycin base is protein bound. Erythromycin is metabolized by hepatic CYP3A4 then excreted in bile. It undergoes intestinal reabsorption. The elimination half life of erythromycin range from 1.5 to 2.5 hours and it is longer in patients with renal impairment.

Indications

Erysol is indicated for the treatment of infections due to susceptible organisms such as in upper and lower respiratory infections, soft tissue infections, skin infections, clostridial infections, urethritis. It is also indicated as an alternative to penicillin in patients who are hypersensitive to penicillin.

Dosage and Administration

To be administered orally. **Erysol** may be taken before or after meals. Children: 30 to 50mg/kg/day or more, in divided doses. The dosage may be doubled in severe infections.

In mild to moderately severe infections the following dosage schedule may be followed:

Bodyweight	Dosage	5 to 10 kg
	2.5 ml 4 times a day or 5ml 12 hourly	
10 to 25 kg	5 ml 4 times a day or 10ml 12 hourly	
Over 25kg	7.5 ml 4 times a day or 15ml 12 hourly	
Adults	10 ml 4 times a day or 20ml 12 hourly	

For very severe or life-threatening infections and infections of the lower respiratory tract these doses may be doubled or increased even more. However, in such instances it may be preferable to initiate treatment by the parenteral route.

The length of therapy will depend on the severity of the infection, and other clinical factors, and varies from 5 to 14 days.

Renal Impairment

Adult with severe renal impairment: a maximum dose of erythromycin 1.5 g daily.

Contraindications

- **Erysol** is contraindicated in patients with known hypersensitivity to this antibiotic.
- Erythromycin is also contraindicated in patients taking astemizole, terfenadine, domperidone, cisapride or pimozone.
- Erythromycin should not be given to patients with a history of QT prolongation (congenital or documented acquired QT prolongation) or ventricular cardiac arrhythmia, including torsades de pointes. (see section *Warnings and Precautions*)
- Erythromycin should not be given to patients with electrolyte disturbances (hypokalaemia, hypomagnesaemia due to the risk of prolongation of QT interval)

Warnings and Precautions

Erysol is principally excreted by the liver and should be used with caution in patients with impaired hepatic function. The safety of **Erysol** for use during pregnancy has not been established.

Rare cases of serious cardiovascular adverse event including deaths, cardiac arrests, torsades de pointes and other ventricular arrhythmia have been observed, when used in patients taking concomitant terfenadine or astemizole.

Prolongation of the QT interval, reflecting effects on cardiac repolarisation imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in patients treated with macrolides including erythromycin (see section *Contraindications and Side Effects*). Fatalities have been reported.

Erythromycin should be used with caution in the following:

- Patients with coronary artery disease, severe cardiac insufficiency, conduction disturbances or clinically relevant bradycardia.
- Patients concomitantly taking other medicinal products associated with QT prolongation (see section *Contraindications*).
- Elderly patients may be more susceptible to drug-associated effects on the QT interval

Epidemiological studies investigating the risk of adverse cardiovascular outcomes with macrolides have shown variable results. Some observational studies have identified a rare short-term risk of arrhythmia, myocardial infarction and cardiovascular mortality associated with macrolides including erythromycin. Consideration of these findings should be balanced with treatment benefits when prescribing erythromycin.

There have been reports of infantile hypertrophic pyloric stenosis (IHPS) occurring in infants following erythromycin therapy. In one cohort of 157 newborns who were given erythromycin for pertussis prophylaxis, seven neonates (5%) developed symptoms of non-bilious vomiting or irritability with feeding and were subsequently diagnosed as having IHPS requiring surgical pyloromyotomy. Since erythromycin may be used in the treatment of conditions in infants which are associated with significant mortality or morbidity (such as pertussis or chlamydia), the benefit of erythromycin therapy needs to be weighed against the potential risk of developing IHPS. Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens- Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalized exanthematous pustulosis (AGEP)], Erysol should be discontinued immediately and appropriate treatment should be urgently initiated.

Infantile hypertrophic pyloric stenosis

There have been reports of infantile hypertrophic pyloric stenosis (IHPS) occurring in infants following erythromycin therapy. Epidemiological studies including data from meta-analyses suggest a 2- to 3-fold increase in the risk of IHPS following exposure to erythromycin in infancy. This risk is highest following exposure to erythromycin during the first 14 days of life. Available data suggests a risk of 2.6% (95% CI: 1.5 - 4.2%) following exposure to erythromycin during this time period. The risk of IHPS in the general population is 0.1 - 0.2%. Since erythromycin may be used in the treatment of conditions in infants which are associated with significant mortality or morbidity (such as pertussis or chlamydia), the benefit of erythromycin therapy needs to be weighed against the potential risk of developing IHPS. Parents should be informed to contact their physician if vomiting or irritability with feeding occurs.

Drug Interaction

The use of erythromycin in patients receiving high doses of theophylline have been associated with an increase of serum theophylline levels and potential theophylline toxicity. In case of theophylline toxicity and/or elevated serum theophylline levels, the dose of theophylline should be reduced while the patient is receiving concomitant erythromycin therapy.

Overdosage

Symptoms include gastrointestinal disturbances, nausea and diarrhoea. Treatment may require gastric lavage and general supportive measures.

Side Effects

The incidence of side effects associated with the use of **Erysol** is low. The more frequent side effects are gastrointestinal, such as abdominal cramping and discomfort, and are dose related. Nausea, vomiting and diarrhoea occur infrequently with the usual oral doses. Prolonged or repeated therapy may produce an overgrowth of non-susceptible bacteria or fungi. If such infections occur, the drug should be discontinued and appropriate therapy instituted.

Allergic reactions ranging from urticaria and mild skin eruptions to anaphylaxis have occurred. There have been isolated reports of reversible hearing loss occurring chiefly in patients with renal insufficiency and in patients receiving high doses of erythromycin.

Cardiac arrest, ventricular fibrillation (frequency not known).

Postmarketing Experience:

Gastrointestinal Disorders: infantile hypertrophic pyloric stenosis. [Skin and Subcutaneous Tissue Disorders](#)

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalized exanthematous pustulosis (AGEP).

Shelf-life

The expiry date is indicated on the packaging.

Presentation

Orange colour suspension when reconstituted, with vanilla flavour.

Erysol For Oral Suspension is supplied in 60 ml ready to dispense bottle.

Reconstitution

Shake the bottle to loosen powder. Add water to the 60ml mark. Shake to wet and disperse powder. Let the suspension stand for 10 to 20 seconds and add a further amount of water to the 60 ml mark. Shake well to disperse powder.

Storage

Store below 30°C. After reconstitution, store in a refrigerator and discard any unused portion after 10 days.

KEEP OUT OF REACH OF CHILDREN JAUHI DARI KANAK-KANAK

For further information, please consult your pharmacist or physician.

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Manufacturer and Product Registration Holder (in Malaysia):

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